



University of Colombo, Sri Lanka

University of Colombo School of Computing

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

BACHELOR OF SCIENCE HONOURS IN COMPUTER SCIENCE

BACHELOR OF SCIENCE HONOURS IN SOFTWARE ENGINEERING

Third Year Examination - Semester I – UCSC AY21 [held in Sept./ Oct. 2024]

SCS3206 — Graph Theory

(Two (2) Hours)

Answer ALL questions



Number of Pages = 16

Number of Questions = 4

To be completed by the candidate

Index Number:

--	--	--	--	--	--	--	--

Important Instructions to candidates:

- I. Students should answer in the medium of English language only using the space provided in this question paper.
- II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- III. Write your index number CLEARLY on each and every page of this Question paper.
- IV. This paper consists of 4 questions in 16 pages (including the Cover Page).
- V. Answer ALL questions.
- VI. Programmable Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are not allowed.
- VII. Non-Programmable calculators are not allowed.
- VIII. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate.

To be completed by the examiners

1	
2	
3	
4	
Total	

The following MCQ questions in **Question 01** and **Question 02** have **05 (five) choices** with one or more correct answers.

Mark the correct choices on the MCQ answer sheet provided.

Question 1

The mark given for a question will vary from **0** (All the incorrect choices are marked & no correct choices are marked) to **+1** (All the correct choices are marked & no incorrect choices are marked).

1. Consider a graph G with 6 vertices and 8 edges. What is the **order** and **size** of the graph?
 - (i) Order = 6, Size = 8
 - (ii) Order = 8, Size = 6
 - (iii) Order = 6, Size = 14
 - (iv) Order = 8, Size = 14
 - (v) Order = 6, Size = 10

2. Which traversal technique(s) visits the nodes in **ascending order** in a **binary search tree**?
 - (i) In-order
 - (ii) Pre-order
 - (iii) Post-order
 - (iv) Level-order
 - (v) Reverse In-order

3. In a simple undirected graph, vertex A has three edges connected to it. What is the **degree** of vertex A ?

(i) 0	(ii) 1	(iii) 3
(iv) 4	(v) 6	

4. Consider a simple undirected graph with 5 vertices and 7 edges. What is the **sum of the degrees** of all the vertices?

(i) 5	(ii) 7	(iii) 10
(iv) 14	(v) 28	

5. What is the **maximum possible degree** of any vertex in a **simple graph** with 5 vertices?
(i) 1 (ii) 4 (iii) 5
(iv) 8 (v) 10
6. In the following graph $G = (V, E)$, which edge(s) is/are **incident** to vertex **C**?
 $V = \{A, B, C, D\}$
 $E = \{(A, B), (A, C), (C, D), (B, D), (A, D)\}$
(i) (A, C) (ii) (C, D) (iii) (A, B)
(iv) (B, D) (v) (A, D)
7. Which of the following statement(s) is/are correct regarding **two adjacent vertices** in a graph?
(i) They have the same degree.
(ii) They are part of the same cycle.
(iii) They have a common neighbor.
(iv) They are connected by a direct edge.
(v) They belong to different components of the graph.
8. Which of the following statement(s) is/are correct regarding a **bipartite graph**?
(i) It has no Euler path
(ii) It contains no cycles
(iii) It contains at least one cycle
(iv) It contains no odd-length cycles
(v) It has an even number of vertices
9. What is the **minimum number of edges** required to make a graph $G = (V, E)$ connected?
(i) V (ii) V-1 (iii) E-1
(iv) E-V+2 (v) V+1
10. Which of the following **graph representation** is most suitable for **dense graphs**?
(i) Adjacency list
(ii) Adjacency matrix
(iii) Incidence list
(iv) Incidence matrix
(v) Path matrix

Question 2

The mark given for a question will vary from **0** (All the incorrect choices are marked & no correct choices are marked) to **+2** (All the correct choices are marked & no incorrect choices are marked).

11. Which of the following statement(s) is/are correct regarding the **incidence matrix** of an undirected graph?
- (i) The incidence matrix has as many rows as the number of vertices.
 - (ii) The incidence matrix has as many columns as the number of edges.
 - (iii) For an undirected graph, the entries of the incidence matrix are either 0, 1, or -1.
 - (iv) The sum of entries in any row of the incidence matrix is always 2 for any vertex.
 - (v) For an undirected graph, each edge will have two 1's in its corresponding column.
12. Which of the following statement(s) is/are correct regarding **two isomorphic graphs**?
- (i) They have the same vertex labels.
 - (ii) They have identical degree sequences.
 - (iii) They have the same number of vertices and edges.
 - (iv) The adjacency relationships between vertices must be preserved.
 - (v) They have different adjacency matrices, depending on how their vertices are labeled.
13. Which of the following is/are **valid degree sequence(s) for a simple graph**?
- (i) (3,3,2,2)
 - (ii) (4,4,4,4,4)
 - (iii) (4,2,6,8,5,1)
 - (iv) (4,3,2,1,5,2,3,6)
 - (v) (3,2,4,1,5,4,3,1)
14. How many **edges** are there in a **forest** consisting of k **trees**, if the forest comprises V **vertices**?
- (i) kV
 - (ii) $V - k$
 - (iii) $V + k$
 - (iv) $kV - k$
 - (v) $kV + k$
15. Which of the following statement(s) is/are correct regarding **trees**?
- (i) Every tree is a bipartite graph.
 - (ii) A tree with n vertices has $n-1$ edges.
 - (iii) A tree can contain cycles if it has more than 3 vertices.
 - (iv) The number of leaves in a tree with n vertices is always $n-1$.
 - (v) In a tree, there is exactly one path between any two vertices.

16. How many edges (E) and vertices (V) are there in a **complete bipartite graph** $K_{m,n}$?
- (i) $E = m * n, V = m * n$
 - (ii) $E = m + n, V = m + n$
 - (iii) $E = m * n, V = m + n$
 - (iv) $E = m + n, V = m * n$
 - (v) $E = m * n - 1, V = m + n$
17. Which of the following statement(s) is/are correct regarding a **cycle graph** and a **wheel graph**?
- (i) A cycle graph C_n has n vertices and n edges.
 - (ii) A cycle graph C_n is always bipartite for all $n \geq 3$.
 - (iii) A cycle graph C_4 is a planar, but a wheel graph W_4 is not.
 - (iv) The degree of the central vertex in a wheel graph W_n is always $n-1$.
 - (v) A wheel graph W_n is formed by connecting an additional vertex to all vertices of a cycle C_{n-1} .
18. Which of the following statement(s) is/are correct regarding **trails, paths, cycles, and circuits**?
- (i) Every cycle is a circuit.
 - (ii) A path is a trail that does not contain a repeated vertex.
 - (iii) A circuit can include repeated edges but no repeated vertices.
 - (iv) A cycle is a trail where the first and last vertices are the same.
 - (v) In a cycle, no vertex or edge can be repeated, except the starting and ending vertex.
19. Which of the following statement(s) is/are correct regarding **Euler trails and Euler circuits**?
- (i) Euler circuit always needs to have an even number of vertices.
 - (ii) Euler trail is a path that visits every vertex of a graph exactly once.
 - (iii) Euler circuit is a closed trail that visits every edge of a graph exactly once.
 - (iv) A connected graph has Euler trail if it has exactly two vertices of odd degree.
 - (v) A connected graph has Euler circuit if and only if all vertices except two have even degrees.
20. Which of the following statement(s) is/are correct regarding **Hamiltonian graphs**?
- (i) A Hamiltonian cycle is a closed path that visits ~~every~~ edge of a graph exactly once.
 - (ii) The existence of a Hamiltonian cycle in a graph can be ~~determined~~ in polynomial time.
 - (iii) A connected graph has a Hamiltonian cycle if and only if every vertex has an even degree.
 - (iv) Every Hamiltonian graph must have exactly two vertices of odd degree, while the rest must have even degrees.
 - (v) A Hamiltonian graph contains a Hamiltonian cycle that visits every vertex exactly once and returns to the starting vertex.

21. Which of the following statement(s) describe(s) **necessary or sufficient conditions** for a graph to be **Hamiltonian**?

- (i) A graph is Hamiltonian if it has no cut vertices.
- (ii) A graph is Hamiltonian if it contains an Eulerian cycle. ~~X~~
- (iii) A graph is Hamiltonian if and only if it contains a Hamiltonian path.
- (iv) If a graph has any vertices with a degree less than 2, it cannot be a Hamiltonian graph.
- (v) A connected graph is Hamiltonian if every vertex has a degree at least $n/2$, where n is the number of vertices in the graph. ✓

22. In the following undirected graph, which of the following statement(s) is/are correct regarding **eccentricity, diameter, and radius**?

$$V = \{A, B, C, D, E\}$$

$$E = \{(A, B), (B, C), (C, D), (D, E), (E, A), (A, C)\}$$

- (i) Vertex C is a central vertex.
- (ii) The radius of the graph is 2.
- (iii) The diameter of the graph is 3.
- (iv) The eccentricity of vertex A is 2.
- (v) The eccentricity of vertex D is 3.

Consider the following adjacency matrix of graph G and answer **question 23** and **24**.

	A	B	C	D	E
A	0	1	0	1	0
B	1	0	1	0	1
C	0	1	0	1	0
D	1	0	1	0	0
E	0	1	0	0	0

23. Which of the following statement(s) is/are correct regarding the given **graph G** ?

- (i) The graph G is a simple graph
- (ii) The graph G is a Hamiltonian graph
- (iii) The graph G contains an Eulerian circuit
- (iv) If we delete vertex E from graph G , the resultant graph will be a cycle graph
- (v) If we delete vertex E from graph G , the resultant graph will be a 2-regular graph

24. Which of the following statement(s) is/are correct regarding **edge connectivity and vertex connectivity** of the given graph G ?

disconnect karnna remove karnna
awshya gaana

- (i) The edge connectivity of the graph G is 2.
- (ii) The vertex connectivity of the graph G is 1.
- (iii) Removing edge (A, B) will disconnect the graph.
- (iv) The edge connectivity is always greater than the vertex connectivity in graph G .
- (v) The minimum degree of G is greater than or equal to the vertex connectivity.

25. Which of the following statement(s) is/are correct regarding **graph operations**?

union
inter
sum - okkoma ekata exist connect
ring - minus intersection
carti-

- (i) The union of two disjoint graphs is always connected.
- (ii) The sum of two graphs G and H is equivalent to their union.
- (iii) The Cartesian product of two connected graphs is always connected.
- (iv) The ring sum of two graphs includes edges that are present in both graphs.
- (v) The intersection of two graphs G and H results in a graph containing only the vertices and edges that are present in both graphs.

26.

Which of the following statement(s) is/are correct regarding **complement graphs**?

- (i) The order of a Graph and its Complement are same.
- (ii) The complement of a complete graph is an empty graph.
- (iii) The complement of a disconnected graph can be connected.
- (iv) The complement of a cycle graph with n vertices is a complete graph with $n-1$ edges.
- (v) The union of graph G and its complement will give a complete graph (K_n) .

27.

Which of the following is/are valid application(s) of **Depth-First Search**?

- (i) Finding bridges and articulation points in a graph.
- (ii) Finding the shortest path in an unweighted graph.
- (iii) Detecting cycles in a directed or undirected graph.
- (iv) Solving mazes or puzzles by exploring all possible paths.
- (v) Finding strongly connected components in a directed graph.

28. Which of the following statement(s) is/are correct regarding **Breadth-First Search**?

- (i) Visit each child node before visiting sibling nodes.
- (ii) Can be used to perform level order traversal of a binary tree.
- (iii) Explore as far as possible along each branch before backtracking.
- (iv) Uses a queue data structure to keep track of nodes to be explored next.
- (v) BFS is more memory-efficient than Depth-First Search for exploring deep trees.

29. Which of the following statement(s) is/are correct regarding **graph coloring**?

- (i) The chromatic number of a wheel graph W_n with $n \geq 3$ is always 3.
- (ii) The edge chromatic number of a cycle graph C_n with $n \geq 3$ is always 3.
- (iii) The edge chromatic number of a path graph P_n with $n \geq 3$ is always 2.
- (iv) When the maximum degree of graph G is 5, its edge chromatic number can be 6.
- (v) The edge chromatic number of a bipartite graph is always equal to its maximum degree of the given bipartite graph.

30. Which of the following statement(s) is/are correct regarding **spanning trees**?

- (i) The number of spanning trees in a complete graph K_5 is 25.
- (ii) Cayley's Theorem can be applied directly to any undirected graph to count the number of spanning trees.
- (iii) A spanning tree is a subgraph that includes any number of vertices (V) of the graph and exactly $V-1$ edges, forming a tree.
- (iv) Kirchhoff's Theorem allows the number of spanning trees to be calculated by taking the determinant of a cofactor of the Laplacian matrix.
- (v) The number of edges that need to be removed from a complete graph K_n to form a spanning tree is $E-V+1$, where E and V are the number of edges and vertices, respectively, in the original graph.

Question 03

A tech company plans to use drones to deliver packages between various distribution centers in a city.

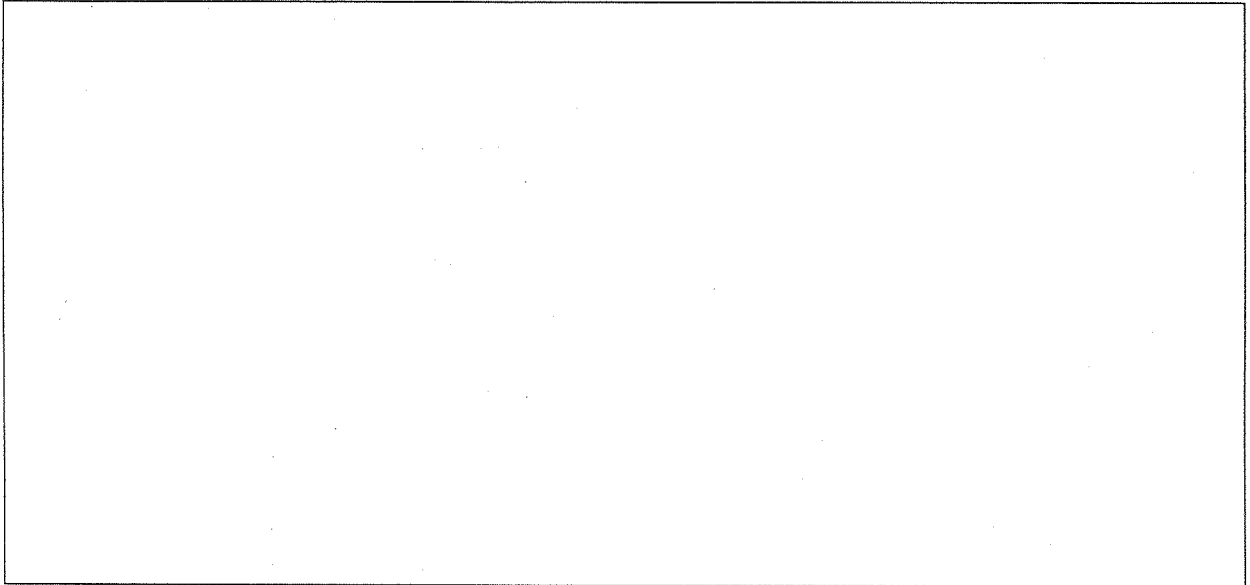
The company has identified **6 distribution centers**, represented as vertices $V = \{A, B, C, D, E, F\}$.

The roads connecting these centers are represented as weighted edges in the graph, where the weights indicate the time (in minutes) for a drone to travel between the centers.

The travel times between the centers are as follows:

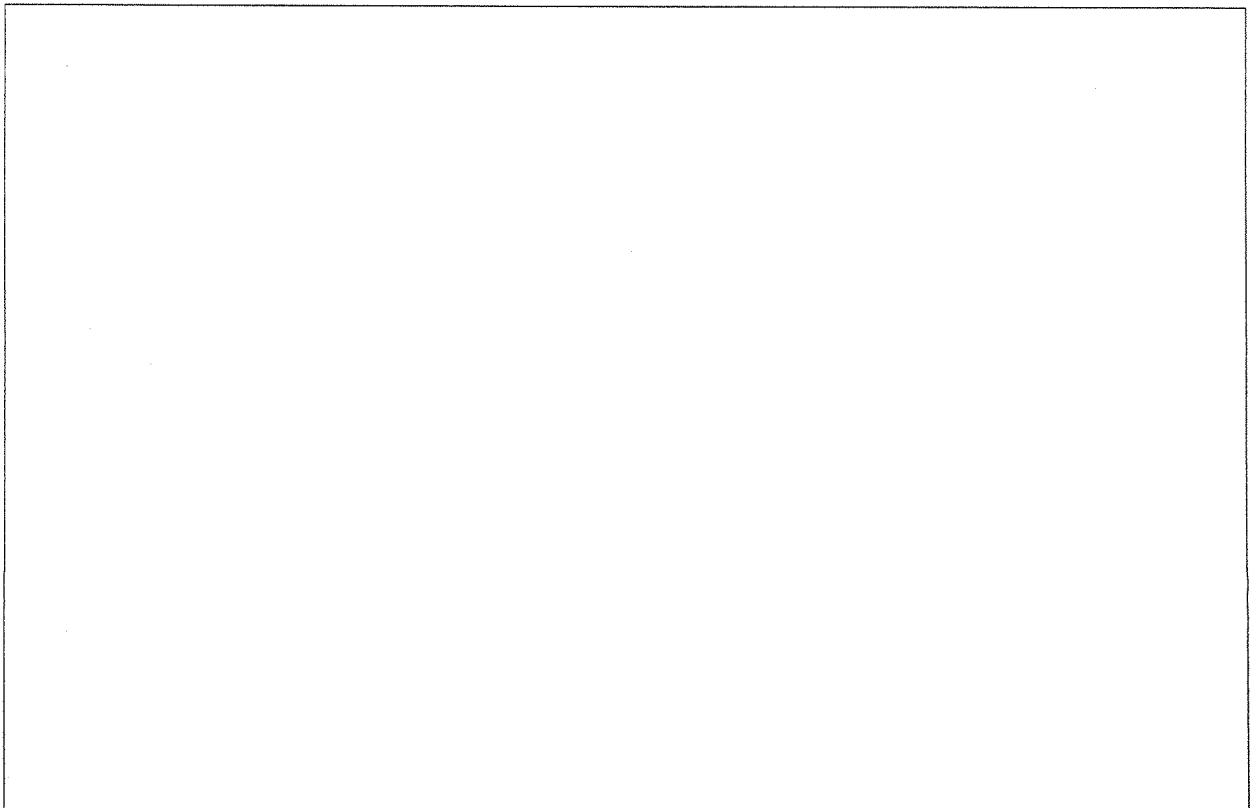
- $A \leftrightarrow B = 10$
- $A \leftrightarrow C = 15$
- $B \leftrightarrow D = 12$
- $C \leftrightarrow D = 8$
- $C \leftrightarrow E = 20$
- $D \leftrightarrow E = 6$
- $E \leftrightarrow F = 9$
- $D \leftrightarrow F = 5$

1. Represent the delivery network as a weighted graph by drawing the graph with vertices and edges labeled with their respective travel times.



[2 marks]

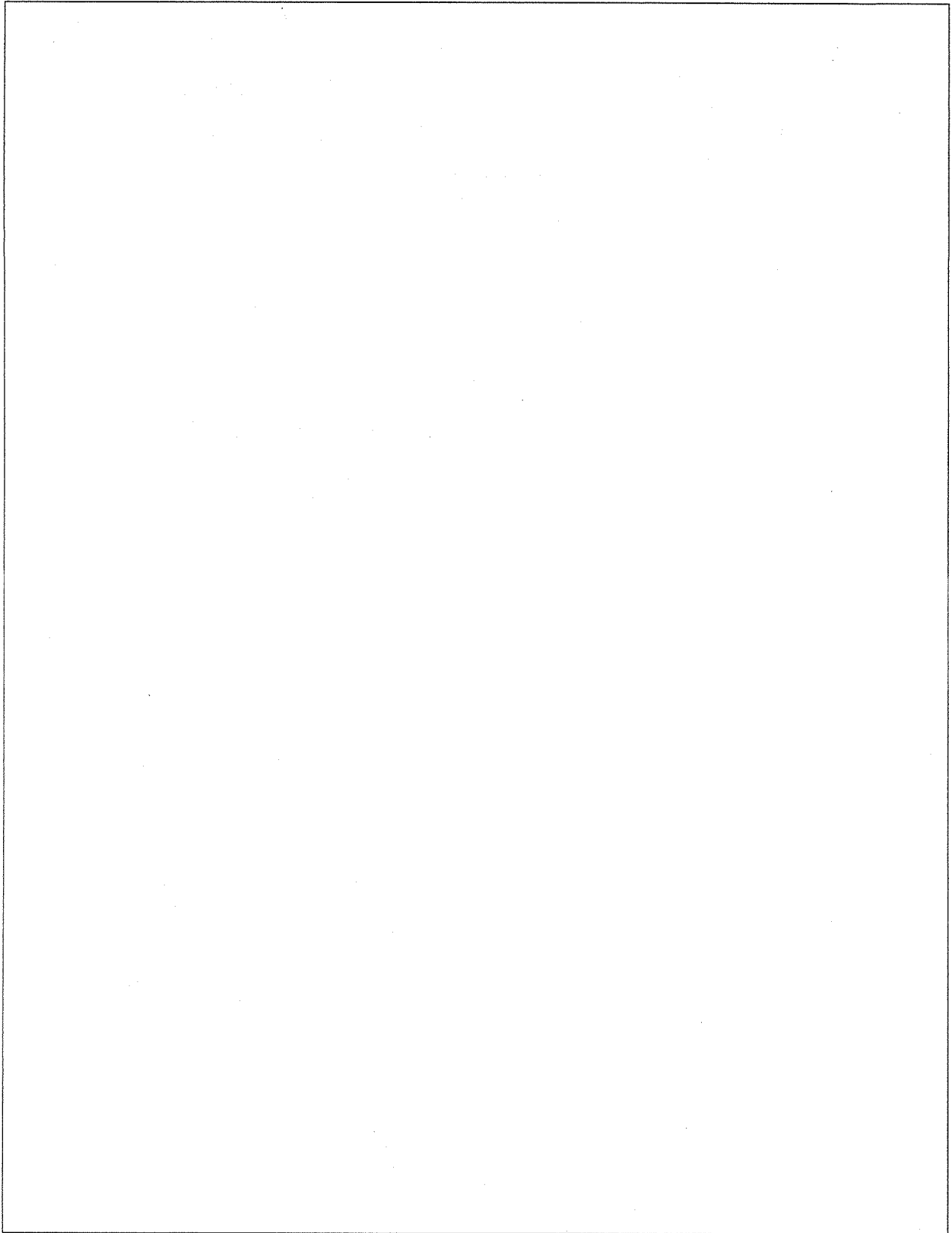
2.
 - (a) List the steps involved in **Prim's Algorithm (Lazy Version)** for constructing a **Minimum Spanning Tree (MST)**.



[4 marks]

Index Number -

- (b) Using **Prim's Algorithm (Lazy Version)**, construct the **Minimum Spanning Tree (MST)** for the given graph, starting from vertex *A*.



[6 marks]

- (c) Find the **distribution center/s** that can be considered as the center of the Minimum Spanning Tree (MST) identified in *Question 3, 2. (b)*. and explain the steps briefly.

[3 marks]

3.

- (a) State the **data structure** used in **Kruskal's Algorithm** when finding the Minimum Spanning Tree (MST).

[2 marks]

- (b) Explain why the chosen data structure is necessary for Kruskal's Algorithm.

[2 marks]

Index Number -

4.

- (a) If the delivery network has been updated with additional distribution centers and new direct routes connecting each distribution center, which of the following algorithm would be the most efficient for finding the Minimum Spanning Tree (MST): *Kruskal's Algorithm*, *Prim's Lazy Version*, or *Prim's Eager Version*?

[2 marks]

- (b) Justify your choice by discussing the characteristics of each algorithm and explain why you chose the particular algorithm and not the other two.

[4 marks]

Question 04

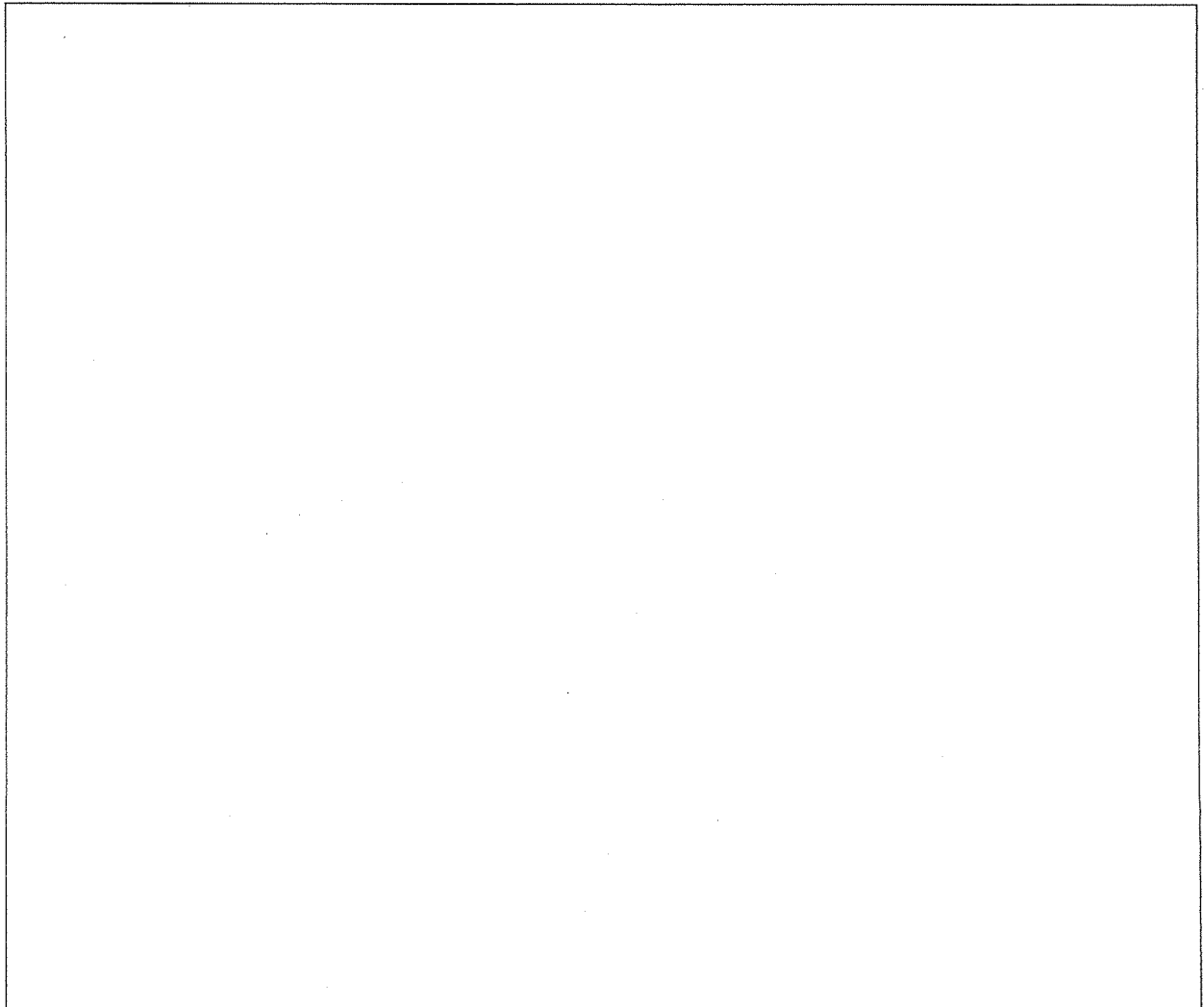
1. In a **cloud computing system**, data needs to be transferred between different servers through network links with limited bandwidth.

The network can be modeled as a **flow network**, where the vertices are servers, and the edges are the data links with capacities representing the maximum amount of data (in GB/s) that can flow through each link.

The following flow network represents the server system:

- Server A (source) can send up to 12 GB/s to server B and 16 GB/s to server C.
- Server B can send 5 GB/s to server C and 10 GB/s to server D.
- Server C can send 5 GB/s to server D and 18 GB/s to server E.
- Server D can send 10 GB/s to server E.
- Server E (sink) is the destination for all data flows.

- (a) Draw the flow network diagram representing the data transfer system.



[2 marks]

Index Number -

- (b) Explain the purpose of incorporating **augmenting paths** and **residual graphs** in network flow models.

[4 marks]

- (c) List the steps of the **Capacity scaling algorithm** used to find the **maximum flow** of a graph.

[4 marks]

Index Number -

- (d) Using the **Capacity Scaling algorithm**, calculate the **maximum flow** from server *A* to server *E*. Provide clear details about the augmenting paths chosen during the process.

[6 marks]

Index Number -

2. For each of the following scenarios, select the most suitable algorithm from the options below and explain why it is the best choice.

Algorithms: *Ford-Fulkerson, Edmonds-Karp, Capacity Scaling*

- (a) You are working with a small transportation network where the number of edges is small, and edge capacities are relatively large.

[3 marks]

- (b) You are working on a large, complex communication network with thousands of nodes and edges, where many of the edges have relatively low capacities, ranging between 1 and 5 units.

[3 marks]

- (c) You are optimizing the flow in a large distributed system where nodes are connected with a mix of high-capacity and low-capacity links. However, you need to avoid bottlenecks and select augmenting paths with the largest available capacities

[3 marks]
